Freudenthal suspension connectivity in homotopy type theory

The Univalent Foundations Program

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Editorial note. The old research item paired the theorem withProof of thm:freudcode1676–1707. This task uses the actual explicitly headedProof of thm:freudenthal1747–1791 and retainsfreudcode as uncounted support.

Editorial note. The author suppresses routine coherence paths at1787–1790. They remain visibly suppressed author steps; no assistant proof completion or formal verification is claimed.

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Allowed standard prerequisites (editorial).

- Explicit univalent foundations boundary. Univalence, function extensionality, path and dependent-pair induction, suspension higher-inductive types, truncations and connected-type induction into truncated families. The wedge lemma and the entire source-local code construction are retained.

Source reference. Original lemma 7.5.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1570–1570. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: lem.

Source reference. Original lemma 8.6.1: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1755–1755. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: lem.

Source reference. Original section 6.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1710–1710. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: section.

Source reference. Original section 8.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1721–1721. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: section.

Source reference. Original corollary 8.4.8: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1754–1754. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: cor.

Source reference. Original corollary 8.8.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1859–1859. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: cor.

Source reference. Original lemma 2.9.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1764–1764. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: lem.

Source reference. Original equation 2.9.4: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1324–1324. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: equation.

Recall that if (A, a_0) and (B, b_0) are pointed types, then their **wedge** $A \vee B$ is defined to be the pushout of $A \xleftarrow{a_0} \mathbf{1} \xrightarrow{b_0} B$. There is a canonical map $i : A \vee B \to A \times B$ defined by the two maps $\lambda a.\,(a, b_0)$ and $\lambda b.\,(a_0, b)$; the following lemma essentially says that this map is highly connected if A and B are so. It is a bit more convenient both to prove and use, however, if we use the characterization of connectedness from original lemma 7.5.7 and substitute in the universal property of the wedge (generalized to type families).

Lemma 8.6.2 (Wedge connectivity lemma)**.** *Suppose that (A, a_0) and (B, b_0) are n- and m-connected pointed types, respectively, with $n, m \geq 0$, and let $P : A \to B \to (n+m)$-*Type*. Then for any $f : \prod_{(a:A)} P(a, b_0)$ and $g : \prod_{(b:B)} P(a_0, b)$ with $p : f(a_0) = g(b_0)$, there exists $h : \prod_{(a:A)} \prod_{(b:B)} P(a, b)$ with homotopies*

$$q : \prod_{a:A} h(a, b_0) = f(a) \qquad \text{and} \qquad r : \prod_{b:B} h(a_0, b) = g(b)$$

such that $p = q(a_0)^{-1} \cdot r(b_0)$.

Proof. Define $Q : A \to \mathcal{U}$ by

$$Q(a) :\equiv \sum_{k:\prod_{(b:B)} P(a,b)} (f(a) = k(b_0)).$$

Then we have $(g, p) : Q(a_0)$. Since $a_0 : \mathbf{1} \to A$ is $(n-1)$-connected, if Q is a family of $(n-1)$-types then we will have $\ell : \prod_{(a:A)} Q(a)$ such that $\ell(a_0) = (g, p)$, in which case we can define $h(a, b) :\equiv \mathsf{pr}_1(\ell(a))(b)$. However, for fixed a, the type $Q(a)$ is the fiber over $f(a)$ of the map

$$\Big(\prod_{b:B} P(a, b)\Big) \to P(a, b_0)$$

given by precomposition with $b_0 : \mathbf{1} \to B$. Since $b_0 : \mathbf{1} \to B$ is $(m-1)$-connected, for this fiber to be $(n-1)$-truncated, by original lemma 8.6.1 it suffices for each type $P(a, b)$ to be an $(n + m)$-type, which we have assumed. □

Let (X, x_0) be a pointed type, and recall the definition of the suspension ΣX from original section 6.5, with constructors $\mathsf{N}, \mathsf{S} : \Sigma X$ and $\mathsf{merid} : X \to (\mathsf{N} = \mathsf{S})$. We regard ΣX as a pointed space with basepoint N, so that we have $\Omega\Sigma X :\equiv (\mathsf{N} =_{\Sigma X} \mathsf{N})$. Then there is a canonical map

$$\sigma : X \to \Omega\Sigma X$$
$$\sigma(x) :\equiv \mathsf{merid}(x) \cdot \mathsf{merid}(x_0)^{-1}.$$

Remark 8.6.3. In classical algebraic topology, one considers the *reduced suspension*, in which the path $\mathsf{merid}(x_0)$ is collapsed down to a point, identifying N and S. The reduced and unreduced suspensions are homotopy equivalent, so the distinction is invisible to our purely homotopy-theoretic eyes — and higher inductive types only allow us to "identify" points up to a higher path anyway, there is no purpose to considering reduced suspensions in homotopy type theory. However, the "unreducedness" of our suspension is the reason for the (possibly unexpected) appearance of $\mathsf{merid}(x_0)^{-1}$ in the definition of σ.

Our goal is now to prove the following.

Theorem 8.6.4 (The Freudenthal suspension theorem)**.** *Suppose that X is n-connected and pointed, with $n \geq 0$. Then the map $\sigma : X \to \Omega\Sigma(X)$ is $2n$-connected.*

We will use the encode-decode method, but applied in a slightly different way. In most cases so far, we have used it to characterize the loop space $\Omega(A, a_0)$ of some type as equivalent to some other type B, by constructing a family $\mathsf{code} : A \to \mathcal{U}$ with $\mathsf{code}(a_0) :\equiv B$ and a family of equivalences $\mathsf{decode} : \prod_{(x:A)} \mathsf{code}(x) \simeq (a_0 = x)$.

In this case, however, we want to show that $\sigma : X \to \Omega\Sigma X$ is $2n$-connected. We could use a truncated version of the previous method, such as we will see in original section 8.7, to prove that $\|X\|_{2n} \to \|\Omega\Sigma X\|_{2n}$ is an equivalence — but this is a slightly weaker statement than the map being $2n$-connected (see original corollarys 8.4.8 and 8.8.5). However, note that in the general case, to prove that $\mathsf{decode}(x)$ is an equivalence, we could equivalently be proving that its fibers are contractible, and we would still be able to use induction over the

base type. This we can generalize to prove connectedness of a map into a loop space, i.e. that the *truncations* of its fibers are contractible. Moreover, instead of constructing code and decode separately, we can construct directly a family of *codes for the truncations of the fibers*.

Definition 8.6.5. If X is n-connected and pointed with $n \geq 0$, then there is a family

$$\mathsf{code} : \prod_{y:\Sigma X} (\mathsf{N} = y) \to \mathcal{U} \tag{8.6.6}$$

such that

$$\mathsf{code}(\mathsf{N}, p) :\equiv \|\mathsf{fib}_\sigma(p)\|_{2n} \equiv \left\| \textstyle\sum_{(x:X)} (\mathsf{merid}(x) \cdot \mathsf{merid}(x_0)^{-1} = p) \right\|_{2n} \tag{8.6.7}$$

$$\mathsf{code}(\mathsf{S}, q) :\equiv \|\mathsf{fib}_{\mathsf{merid}}(q)\|_{2n} \equiv \left\| \textstyle\sum_{(x:X)} (\mathsf{merid}(x) = q) \right\|_{2n}. \tag{8.6.8}$$

Our eventual goal will be to prove that $\mathsf{code}(y, p)$ is contractible for all $y : \Sigma X$ and $p : \mathsf{N} = y$. Applying this with $y :\equiv \mathsf{N}$ will show that all fibers of σ are $2n$-connected, and thus σ is $2n$-connected.

Proof of Definition 8.6.5. We define $\mathsf{code}(y, p)$ by induction on $y : \Sigma X$, where the first two cases are (8.6.7) and (8.6.8). It remains to construct, for each $x_1 : X$, a dependent path

$$\mathsf{code}(\mathsf{N}) =^{\lambda y.\, (\mathsf{N}=y)\to\mathcal{U}}_{\mathsf{merid}(x_1)} \mathsf{code}(\mathsf{S}).$$

By original lemma 2.9.6, this is equivalent to giving a family of paths

$$\prod_{q:\mathsf{N}=\mathsf{S}} \mathsf{code}(\mathsf{N})(\mathsf{transport}^{\lambda y.\, (\mathsf{N}=y)}(\mathsf{merid}(x_1)^{-1}, q)) = \mathsf{code}(\mathsf{S})(q).$$

And by univalence and transport in path types, this is equivalent to a family of equivalences

$$\prod_{q:\mathsf{N}=\mathsf{S}} \mathsf{code}(\mathsf{N}, q \cdot \mathsf{merid}(x_1)^{-1}) \simeq \mathsf{code}(\mathsf{S}, q).$$

We will define a family of maps

$$\prod_{q:\mathsf{N}=\mathsf{S}} \mathsf{code}(\mathsf{N}, q \cdot \mathsf{merid}(x_1)^{-1}) \to \mathsf{code}(\mathsf{S}, q). \tag{8.6.9}$$

and then show that they are all equivalences. Thus, let $q : \mathsf{N} = \mathsf{S}$; by the universal property of truncation and the definitions of $\mathsf{code}(\mathsf{N}, -)$ and $\mathsf{code}(\mathsf{S}, -)$, it will suffice to define for each $x_2 : X$, a map

$$\left(\mathsf{merid}(x_2) \cdot \mathsf{merid}(x_0)^{-1} = q \cdot \mathsf{merid}(x_1)^{-1}\right) \to \left\| \textstyle\sum_{(x:X)} (\mathsf{merid}(x) = q) \right\|_{2n}.$$

Now for each $x_1, x_2 : X$, this type is $2n$-truncated, while X is n-connected. Thus, by Lemma 8.6.2, it suffices to define this map when x_1 is x_0, when x_2 is x_0, and check that they agree when both are x_0.

When x_1 is x_0, the hypothesis is $r : \mathsf{merid}(x_2) \cdot \mathsf{merid}(x_0)^{-1} = q \cdot \mathsf{merid}(x_0)^{-1}$. Thus, by canceling $\mathsf{merid}(x_0)^{-1}$ from r to get $r' : \mathsf{merid}(x_2) = q$, so we can define the image to be $|(x_2, r')|_{2n}$.

When x_2 is x_0, the hypothesis is $r : \mathsf{merid}(x_0) \cdot \mathsf{merid}(x_0)^{-1} = q \cdot \mathsf{merid}(x_1)^{-1}$. Rearranging this, we obtain $r'' : \mathsf{merid}(x_1) = q$, and we can define the image to be $|(x_1, r'')|_{2n}$.

Finally, when both x_1 and x_2 are x_0, it suffices to show the resulting r' and r'' agree; this is an easy lemma about path composition. This completes the definition of (8.6.9). To show that it is a family of equivalences, since being an equivalence is a mere proposition and $x_0 : \mathbf{1} \to X$ is (at least) (-1)-connected, it suffices to assume x_1 is x_0. In this case, inspecting the above construction we see that it is essentially the $2n$-truncation of the function that cancels $\mathsf{merid}(x_0)^{-1}$, which is an equivalence. □

In addition to (8.6.7) and (8.6.8), we will need to extract from the construction of code some information about how it acts on paths. For this we use the following lemma.

Lemma 8.6.10. *Let $A : \mathcal{U}$, $B : A \to \mathcal{U}$, and $C : \prod_{(a:A)} B(a) \to \mathcal{U}$, and also $a_1, a_2 : A$ with $m : a_1 = a_2$ and $b : B(a_2)$. Then the function*

$$\widehat{\mathsf{transport}^C}(\mathsf{pair}^=(m, t), -) : C(a_1, \mathsf{transport}^B(m^{-1}, b)) \to C(a_2, b),$$

where $t : \mathsf{transport}^B(m, \mathsf{transport}^B(m^{-1}, b)) = b$ is the obvious coherence path and $\widehat{C} : (\sum_{(a:A)} B(a)) \to \mathcal{U}$ is the uncurried form of C, is equal to the equivalence obtained by univalence from the composite

$$\begin{aligned} C(a_1, \mathsf{transport}^B(m^{-1}, b)) &= \mathsf{transport}^{\lambda a.\, B(a) \to \mathcal{U}}(m, C(a_1))(b) && \text{(by (2.9.4))} \\ &= C(a_2, b). && (\text{by } \mathsf{happly}(\mathsf{apd}_C(m), b)) \end{aligned}$$

Proof. By path induction, we may assume a_2 is a_1 and m is refl_{a_1}, in which case both functions are the identity. □

We apply this lemma with $A :\equiv \Sigma X$ and $B :\equiv \lambda y.\, (\mathsf{N} = y)$ and $C :\equiv \mathsf{code}$, while $a_1 :\equiv \mathsf{N}$ and $a_2 :\equiv \mathsf{S}$ and $m :\equiv \mathsf{merid}(x_1)$ for some $x_1 : X$, and finally $b :\equiv q$ is some path $\mathsf{N} = \mathsf{S}$. The computation rule for induction over ΣX identifies $\mathsf{apd}_C(m)$ with a path constructed in a certain way out of univalence and function extensionality. The second function described in Lemma 8.6.10 essentially consists of undoing these applications of univalence and function extensionality, reducing back to the particular functions (8.6.9) that we defined using Lemma 8.6.2. Therefore, Lemma 8.6.10 says that transporting along $\mathsf{pair}^=(q, t)$ essentially recovers these functions.

Finally, by construction, when x_1 or x_2 coincides with x_0 and the input is in the image of $|-|_{2n}$, we know more explicitly what these functions are. Thus, for any $x_2 : X$, we have

$$\mathsf{transport}^{\widehat{\mathsf{code}}}(\mathsf{pair}^=(\mathsf{merid}(x_0), t), |(x_2, r)|_{2n}) = |(x_1, r')|_{2n} \tag{8.6.11}$$

where $r : \mathsf{merid}(x_2) \cdot \mathsf{merid}(x_0)^{-1} = \mathsf{transport}^B(\mathsf{merid}(x_0)^{-1}, q)$ is arbitrary as before, and $r' : \mathsf{merid}(x_2) = q$ is obtained from r by identifying its end point with $q \cdot \mathsf{merid}(x_0)^{-1}$ and canceling $\mathsf{merid}(x_0)^{-1}$. Similarly, for any $x_1 : X$, we have

$$\mathsf{transport}^{\widehat{\mathsf{code}}}(\mathsf{pair}^=(\mathsf{merid}(x_1), t), |(x_0, r)|_{2n}) = |(x_1, r'')|_{2n} \tag{8.6.12}$$

where $r : \mathsf{merid}(x_0) \cdot \mathsf{merid}(x_0)^{-1} = \mathsf{transport}^B(\mathsf{merid}(x_1)^{-1}, q)$, and $r'' : \mathsf{merid}(x_1) = q$ is obtained by identifying its end point and rearranging paths.

Proof of Theorem 8.6.4. It remains to show that $\mathsf{code}(y,p)$ is contractible for each $y : \Sigma X$ and $p : \mathsf{N} = y$. First we must choose a center of contraction, say $c(y,p) : \mathsf{code}(y,p)$. This corresponds to the definition of the function encode in our previous proofs, so we define it by transport. Note that in the special case when y is N and p is $\mathsf{refl}_{\mathsf{N}}$, we have

$$\mathsf{code}(\mathsf{N}, \mathsf{refl}_{\mathsf{N}}) \equiv \left\| \textstyle\sum_{(x:X)} (\mathsf{merid}(x) \cdot \mathsf{merid}(x_0)^{-1} = \mathsf{refl}_{\mathsf{N}}) \right\|_{2n}.$$

Thus, we can choose $c(\mathsf{N}, \mathsf{refl}_{\mathsf{N}}) :\equiv \left|(x_0, \mathsf{rinv}_{\mathsf{merid}(x_0)})\right|_{2n}$, where rinv_q is the obvious path $q \cdot q^{-1} = \mathsf{refl}$ for any q. We can now obtain $c : \prod_{(y:\Sigma X)} \prod_{(p:\mathsf{N}=y)} \mathsf{code}(y,p)$ by path induction on p, but it will be important below that we can also give a concrete definition in terms of transport:

$$c(y,p) :\equiv \mathsf{transport}^{\hat{\mathsf{code}}}(\mathsf{pair}^{=}(p, \mathsf{tid}_p), c(\mathsf{N}, \mathsf{refl}_{\mathsf{N}}))$$

where $\hat{\mathsf{code}} : \left(\sum_{(y:\Sigma X)} (\mathsf{N} = y)\right) \to \mathcal{U}$ is the uncurried version of code, and $\mathsf{tid}_p : p_*(\mathsf{refl}) = p$ is a standard lemma.

Next, we must show that every element of $\mathsf{code}(y,p)$ is equal to $c(y,p)$. Again, by path induction, it suffices to assume y is N and p is $\mathsf{refl}_{\mathsf{N}}$. In fact, we will prove it more generally when y is N and p is arbitrary. That is, we will show that for any $p : \mathsf{N} = \mathsf{N}$ and $d : \mathsf{code}(\mathsf{N}, p)$ we have $d = c(\mathsf{N}, p)$. Since this equality is a $(2n-1)$-type, we may assume d is of the form $|(x_1, r)|_{2n}$ for some $x_1 : X$ and $r : \mathsf{merid}(x_1) \cdot \mathsf{merid}(x_0)^{-1} = p$.

Now by a further path induction, we may assume that r is reflexivity, and p is $\mathsf{merid}(x_1) \cdot \mathsf{merid}(x_0)^{-1}$. (This is why we generalized to arbitrary p above.) Thus, we have to prove that

$$\left|(x_1, \mathsf{refl}_{\mathsf{merid}(x_1) \cdot \mathsf{merid}(x_0)^{-1}})\right|_{2n} \;=\; c\left(\mathsf{N}, \mathsf{refl}_{\mathsf{merid}(x_1) \cdot \mathsf{merid}(x_0)^{-1}}\right). \qquad (8.6.13)$$

By definition, the right-hand side of this equality is

$$\begin{aligned}
&\mathsf{transport}^{\hat{\mathsf{code}}}\Big(\mathsf{pair}^{=}(\mathsf{merid}(x_1) \cdot \mathsf{merid}(x_0)^{-1}, _),\ |(x_0, _)|_{2n}\Big) \\
&\quad = \mathsf{transport}^{\hat{\mathsf{code}}}\Big(\mathsf{pair}^{=}(\mathsf{merid}(x_0)^{-1}, _), \\
&\qquad\qquad \mathsf{transport}^{\hat{\mathsf{code}}}\Big(\mathsf{pair}^{=}(\mathsf{merid}(x_1), _),\ |(x_0, _)|_{2n}\Big)\Big) \\
&\quad = \mathsf{transport}^{\hat{\mathsf{code}}}\Big(\mathsf{pair}^{=}(\mathsf{merid}(x_0)^{-1}, _),\ |(x_1, _)|_{2n}\Big) = |(x_1, _)|_{2n}
\end{aligned}$$

where the underscore _ ought to be filled in with suitable coherence paths. Here the first step is functoriality of transport, the second invokes (8.6.12), and the third invokes (8.6.11) (with transport moved to the other side). Thus we have the same first component as the left-hand side of (8.6.13). We leave it to the reader to verify that the coherence paths all cancel, giving reflexivity in the second component. □